Cisco PE-15 KR Puram Access Routing Protocol Flap Resolution SOP

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_8083

Description

The Cisco PE-15 at the KR Puram Access site is a critical node in the network, responsible for maintaining BGP, OSPF, and ISIS routing protocols. The failure domain of Routing Protocol Flap is a complex issue that can occur due to various reasons such as BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This SOP is designed to troubleshoot and resolve the issue in a timely and efficient manner. The Cisco PE-15 is a high-capacity router that utilizes a distributed architecture to provide high availability and scalability. The BGP and OSPF protocols are used to exchange routing information between the PE-15 and other routers in the network, while the ISIS protocol is used to provide a loop-free topology. The failure of any of these protocols can lead to a routing protocol flap, resulting in packet loss and network instability. The KR Puram Access site is a critical location that requires a stable and high-performance network connection. Therefore, it is essential to identify and resolve the root cause of the issue as quickly as possible to minimize downtime and ensure business continuity.

Pre-requisite Checks

- Verify that all redundant links are operational and functioning correctly.

- Ensure that the network team has been dispatched to the KR Puram Access site to perform any necessary physical checks and repairs.

- Check the network management system (NMS) for any alarms or notifications related to the PE-15 or the KR Puram Access site.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 123, Description: This command displays a summary of the BGP routing table, including the number of routes and the number of peers. It helps to identify any BGP route churn or issues with the BGP protocol.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the
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OSPF neighbors, including their state and dead time. It helps to
identify any OSPF adjacency loss or issues with the OSPF protocol.

Command: show isis database detail, Expected Output: IS-IS Level-1
Database, Description: This command displays the ISIS database,
including the LSPs and their sequence numbers. It helps to identify
any ISIS LSP flooding or issues with the ISIS protocol.
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Resolution Steps

Step 1: Step: Clear BGP Session, Command: clear ip bgp *, Description: This command clears the BGP session, including the routing table and the BGP peers. It helps to resolve BGP route churn and issues with the BGP protocol.

Step 2: Step: Rebuild OSPF Adjacency, Command: clear ip ospf process, Description: This command rebuilds the OSPF adjacency, including the OSPF neighbors and the OSPF routing table. It helps to resolve OSPF adjacency loss and issues with the OSPF protocol.

Step 3: Step: Flush ISIS LSPs, Command: clear isis database, Description: This command flushes the ISIS LSPs, including the LSPs and their sequence numbers. It helps to resolve ISIS LSP flooding and issues with the ISIS protocol.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is
123, Description: This command verifies that the BGP routing table has
been updated and that the BGP protocol is functioning correctly.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command verifies that
the OSPF neighbors have been rebuilt and that the OSPF protocol is
functioning correctly.

Command: show isis database detail, Expected Output: IS-IS Level-1
Database, Description: This command verifies that the ISIS database
has been updated and that the ISIS protocol is functioning correctly.
```

Roll-back Procedures

- Step: Restore BGP Session, Command: copy running-config startup-config, Description: This command restores the BGP session, including the routing table and the BGP peers. It helps to revert any changes made during the troubleshooting process.

- Step: Rebuild OSPF Adjacency, Command: copy running-config startup-config, Description: This command rebuilds the OSPF adjacency, including the OSPF neighbors and the OSPF routing table. It helps to revert any changes made during the troubleshooting process.

- Step: Flush ISIS LSPs, Command: copy running-config startup-config, Description: This command flushes the ISIS LSPs, including the LSPs and their sequence numbers. It helps to revert any changes made during the troubleshooting process.